STAT3888 Visualisations and Tables for Question 2

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To Do List

Tables.

- Population characteristics table for each variable
  - Categorical format: number of observations (proportion)
  - Numerical format: mean (range)
- Three models; coefficient of `ExerciseGuidelines` (and interaction effects if applicable), and p-value. Models: -crude model (no confounders)
  - full model
  - full model + interaction with `SmokingDuration` and `AlcoholRisk`
- Table of actual odds ratios and confidence intervals for `ExerciseGuidelines` for both `ALT3` and `GGT3` full models

Plots.

- `ExerciseGuidelines` (x-axis) against (possibly numeric) `ALT` and `GGT` (y-axis) respectively, or facet-wrapped barplot with significance levels and error bars of SEs
- Interaction effect plots

Tables

The tables here are not the final versions and I will make them using LaTeX `booktabs` instead on a separate LaTeX document.

▼ Code

```
# Initial setup
knitr::opts_chunk$set(warning = FALSE, message = FALSE,
                      fig.align='center', fig.dim=c(10, 8))

# Libraries
library(tidyverse)
library(here)
library(knitr)
```

```
library(kableExtra)
library(nnet)
library(stargazer)
library(broom)
library(ggpubr)
library(emmeans)
library(scales)

# Data
dat.full = readRDS(here("Objects", "dat_full.rds"))
```

We make first the population characteristics table, using `ALT3` and `GGT3` as the ALT and GGT variables. Then, we have:

- Numerical variables: `Age` and `Waist.`
- Categorical Variables: `AlcoholRisk`, `SmokingDuration`, `ExerciseGuidelines`, `Sex`, `NonShoolQuals`, `IncomeDecile`, `ALT3`, `GGT3.`

Tables 1: Variable Definitions

Variable	Meaning	Type / coding
`ALT3`	Alanine aminotransferase (ALT) category	Ordered categorical factor: Low vs Borderline vs High
`GGT3`	Gamma-glutamyl transferase (GGT) category	Ordered categorical factor: Low vs Borderline vs High
`ExerciseGuidelines`	Whether recommended physical-activity guidelines were met	Binary factor: Yes / No
`Age`	Age in years	Continuous numeric
`Sex`	Sex	Binary factor: Female / Male
`Waist`	Waist circumference in cm	Continuous numeric
`AlcoholRisk`	Alcohol consumption risk level	Binary factor: Low risk / Medium–High risk

`SmokingDuration`	Duration of smoking history	Ordered categorical factor, grouped by years smoked
`NonSchoolQuals`	Whether a non-school educational qualification was completed	Binary factor: Yes / No
`IncomeDecile`	Income decile	Ordered categorical factor, deciles 1–10

Tables 2: Population Characteristics Summary

Again, note that this is not the final tables, and are just drafts for the final ones on LaTeX.

▼ Code

```
# Numerical summaries: mean, and IQR
age.summary = summary(dat.full$Age)[c(4, 2, 5)]
waist.summary = summary(dat.full$Waist)[c(4, 2, 5)]
num.summary = rbind(age.summary, waist.summary) %>%
  as.data.frame() %>%
  rename("Q1" = "1st Qu.", "Q3" = "3rd Qu.")
rownames(num.summary) = c("Age", "Waist")
num.summary %>% kable(caption = "Numerical Summary")
```

Numerical Summary

	Mean	Q1	Q3
Age	51.75315	37.000	62
Waist	94.35760	83.025	104

▼ Code

```
# Categorical Summary: Variable | Class | Count | Proportion
categoricals = c("AlcoholRisk", "SmokingDuration", "ExerciseGui
                 "NonSchoolQuals", "IncomeDecile", "ALT3", "GGT
cat.summary = map_df(categoricals, function(var) {
  dat.full %>%
    group_by(Class = as.character(.data[[var]])) %>%
    summarise(Count = n(), .groups = 'drop') %>%
```

```
    mutate(Variable = var, Proportion = round(Count / sum(Count
    select(Variable, Class, Count, Proportion)
})

cat.summary %>% kable(caption = "Categorical Summary Table")
```

Categorical Summary Table

Variable	Class	Count	Proportion
AlcoholRisk	Low risk	4155	87.3
AlcoholRisk	Medium/High risk	607	12.7
SmokingDuration	10 to 19 years	556	11.7
SmokingDuration	20 to 29 years	479	10.1
SmokingDuration	30 to 39 years	352	7.4
SmokingDuration	40 to 49 years	238	5.0
SmokingDuration	50 years and over	87	1.8
SmokingDuration	Less than 10 years	512	10.8
SmokingDuration	Never smoked daily	2538	53.3
ExerciseGuidelines	No	2298	48.3
ExerciseGuidelines	Yes	2464	51.7
Sex	Female	2519	52.9
Sex	Male	2243	47.1
NonSchoolQuals	No	1681	35.3
NonSchoolQuals	Yes	3081	64.7
IncomeDecile	1	301	6.3
IncomeDecile	10	487	10.2
IncomeDecile	2	473	9.9
IncomeDecile	3	539	11.3
IncomeDecile	4	560	11.8
IncomeDecile	5	448	9.4
IncomeDecile	6	448	9.4
IncomeDecile	7	478	10.0
IncomeDecile	8	482	10.1
IncomeDecile	9	546	11.5
ALT3	Borderline	525	11.0

ALT3	High	579	12.2
ALT3	Low	3658	76.8
GGT3	Borderline	268	5.6
GGT3	High	624	13.1
GGT3	Low	3870	81.3

Maybe for the categorical one we can split them into the main variables (`ALT3`/`GGT3` and `ExerciseGuidelines` only), and in the appendix we have the proportions of everything else.

▼ Code

```
# Main variables table
main.cat = c("ExerciseGuidelines", "ALT3", "GGT3")
cat.summary = map_df(main.cat, function(var) {
  dat.full %>%
    group_by(Class = as.character(.data[[var]])) %>%
    summarise(Count = n(), .groups = 'drop') %>%
    mutate(Variable = var, Proportion = round(Count / sum(Count
    select(Variable, Class, Count, Proportion)
})

cat.summary %>% kable(caption = "Main Categorical Summary Table'
```

Main Categorical Summary Table

Variable	Class	Count	Proportion
ExerciseGuidelines	No	2298	48.3
ExerciseGuidelines	Yes	2464	51.7
ALT3	Borderline	525	11.0
ALT3	High	579	12.2
ALT3	Low	3658	76.8
GGT3	Borderline	268	5.6
GGT3	High	624	13.1
GGT3	Low	3870	81.3

▼ Code

```
# Other variables table
other.cat = c("AlcoholRisk", "SmokingDuration", "Sex",
              "NonSchoolQuals", "IncomeDecile")
```

```
cat.summary = map_df(other.cat, function(var) {
  dat.full %>%
    group_by(Class = as.character(.data[[var]])) %>%
    summarise(Count = n(), .groups = 'drop') %>%
    mutate(Variable = var, Proportion = round(Count / sum(Count
    select(Variable, Class, Count, Proportion)
})

cat.summary %>% kable(caption = "Other Categorical Summary Table
```

Other Categorical Summary Table

Variable	Class	Count	Proportion
AlcoholRisk	Low risk	4155	87.3
AlcoholRisk	Medium/High risk	607	12.7
SmokingDuration	10 to 19 years	556	11.7
SmokingDuration	20 to 29 years	479	10.1
SmokingDuration	30 to 39 years	352	7.4
SmokingDuration	40 to 49 years	238	5.0
SmokingDuration	50 years and over	87	1.8
SmokingDuration	Less than 10 years	512	10.8
SmokingDuration	Never smoked daily	2538	53.3
Sex	Female	2519	52.9
Sex	Male	2243	47.1
NonSchoolQuals	No	1681	35.3
NonSchoolQuals	Yes	3081	64.7
IncomeDecile	1	301	6.3
IncomeDecile	10	487	10.2
IncomeDecile	2	473	9.9
IncomeDecile	3	539	11.3
IncomeDecile	4	560	11.8
IncomeDecile	5	448	9.4
IncomeDecile	6	448	9.4
IncomeDecile	7	478	10.0
IncomeDecile	8	482	10.1
IncomeDecile	9	546	11.5

Then the other categorical variables will be made into a table with the numerical ones instead.

Tables 3: Models

I think it is a bit difficult to combine the crude/main models with the interaction model, since we are comparing different things when we compare `ExerciseGuidelines`, so we split these into two tables. Additionally, I also present the table using odds ratios as well. This way, the first answers whether adjustment change the apparent exercise association; the second table (and maybe figures later on) could answer whether that association differ according to alcohol risk or smoking duration.

▼ Code

```
# ALT Model fits
ALT.crude = nnet::multinom(formula = ALT3 ~ ExerciseGuidelines,
                           data = dat.full, Hess = TRUE)
```

```
# weights:  9 (4 variable)
initial  value 5231.591719
final  value 3337.314874
converged
```

▼ Code

```
ALT.main = nnet::multinom(formula = ALT3 ~ ExerciseGuidelines +
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, Hess = TRUE)
```

```
# weights:  69 (44 variable)
initial  value 5231.591719
iter  10 value 3332.778645
iter  20 value 3113.516122
iter  30 value 3083.159558
iter  40 value 3082.606611
final  value 3082.585935
converged
```

▼ Code

```
ALT.int = nnet::multinom(formula = ALT3 ~ ExerciseGuidelines + A
    SmokingDuration + NonSchoolQuals + IncomeDecile + Sex + Age
      ExerciseGuidelines : AlcoholRisk + ExerciseGuidelines : Sr
```

```
    data = dat.full, Hess = TRUE)
```

```
# weights:  90 (58 variable)
initial  value 5231.591719
iter  10 value 3315.909335
iter  20 value 3117.558534
iter  30 value 3073.457298
iter  40 value 3072.360529
iter  50 value 3072.334952
final  value 3072.333507
converged
```

▼ Code

```
# GGT Model fits
GGT.crude = nnet::multinom(formula = GGT3 ~ ExerciseGuidelines,
                           data = dat.full, Hess = TRUE)
```

```
# weights:  9 (4 variable)
initial  value 5231.591719
iter  10 value 2822.049929
iter  10 value 2822.049928
final  value 2822.049928
converged
```

▼ Code

```
GGT.main = nnet::multinom(formula = GGT3 ~ ExerciseGuidelines +
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, Hess = TRUE)
```

```
# weights:  69 (44 variable)
initial  value 5231.591719
iter  10 value 3091.392369
iter  20 value 2669.879096
iter  30 value 2635.855949
iter  40 value 2634.622471
iter  50 value 2634.533708
final  value 2634.532474
converged
```

▼ Code

```
GGT.int = nnet::multinom(formula = GGT3 ~ ExerciseGuidelines + A
    SmokingDuration + NonSchoolQuals + IncomeDecile + Sex + Age
```

```
      ExerciseGuidelines : AlcoholRisk + ExerciseGuidelines : Sr
    data = dat.full, Hess = TRUE)
```

```
# weights:  90 (58 variable)
initial  value 5231.591719
iter  10 value 3097.749588
iter  20 value 2685.304961
iter  30 value 2632.472047
iter  40 value 2630.861588
iter  50 value 2630.714576
iter  60 value 2630.697693
final  value 2630.697446
converged
```

▼ Code

```
# Summary of just the ExerciseGuidelines coefficient + p-value
extract_exercise = function(model, model_name) {

  tidy(model) %>% filter(grepl("^ExerciseGuidelines", term)) %>%
    mutate(Model = model_name) %>%
    select(Model, Outcome = y.level, term, estimate, std.error,

}
ALT.exercise = bind_rows(extract_exercise(ALT.crude, "Crude"),
                         extract_exercise(ALT.main,  "Adjusted")

GGT.exercise = bind_rows(extract_exercise(GGT.crude, "Crude"),
                         extract_exercise(GGT.main,  "Adjusted")

# ALT Table
ALT.exercise.table = ALT.exercise %>%
  mutate(estimate = round(estimate, 3),
         p.value = case_when(p.value < 0.0001 ~ "<0.0001",
                             TRUE ~ sprintf("%.4f", p.value))) %
  select(Outcome, Model, estimate, p.value) %>%
  pivot_wider(names_from = Model, values_from = c(estimate, p.va
              names_glue = "{Model}_{.value}")

kable(ALT.exercise.table, caption = "Exercise Guidelines and ALT
      align = "c")
```

Exercise Guidelines and ALT Category

Outcome	Crude_estimate	Adjusted_estimate	Crude_p.value	Adjusted_p.value

Borderline	-0.088	-0.120	0.3495	0.2289
High	0.262	0.106	0.0036	0.2727

▼ Code

```
# GGT Table
GGT.exercise.table = GGT.exercise %>%
  mutate(estimate = round(estimate, 3),
         p.value = case_when(p.value < 0.0001 ~ "<0.0001",
                             TRUE ~ sprintf("%.4f", p.value))) %
  select(Outcome, Model, estimate, p.value) %>%
  pivot_wider(names_from = Model, values_from = c(estimate, p.va
              names_glue = "{Model}_{.value}")

kable(GGT.exercise.table, caption = "Exercise Guidelines and GG
      align = "c")
```

Exercise Guidelines and GGT Category

Outcome	Crude_estimate	Adjusted_estimate	Crude_p.value	Adjusted_p.value
Borderline	0.243	0.126	0.0549	0.3409
High	0.536	0.300	<0.0001	0.0014

▼ Code

```
# Summary using Odds Ratios
extract_exercise_OR = function(model, model_name) {

  tidy(model, conf.int = TRUE, exponentiate = TRUE) %>%
    filter(grepl("^ExerciseGuidelines", term)) %>%
    mutate(Model = model_name) %>%
    select(Model, Outcome = y.level, term, OR = estimate, conf.
           p.value)

}

# ALT
ALT.exercise.OR = bind_rows(extract_exercise_OR(ALT.crude, "Cru
                            extract_exercise_OR(ALT.main,  "Adj
ALT.exercise.OR = ALT.exercise.OR %>%
  mutate(`OR (95% CI)` = sprintf("%.2f (%.2f, %.2f)", OR, conf.
         `p-value` = ifelse(p.value < 0.0001, "<0.0001", sprint
kable(ALT.exercise.OR,
      caption = "Exercise Guidelines and ALT Category (Odds Rat
```

Exercise Guidelines and ALT Category (Odds Ratios)

Model	Outcome	term	OR	conf.low	conf.high	p.value	C (9 C
Crude	Borderline	ExerciseGuidelinesNo	0.9160768	0.7624101	1.100715	0.3494507	0. (0 1.
Crude	High	ExerciseGuidelinesNo	1.2990048	1.0894832	1.548820	0.0035576	1. (1 1.
Adjusted	Borderline	ExerciseGuidelinesNo	0.8873524	0.7303697	1.078076	0.2289214	0. (0 1.
Adjusted	High	ExerciseGuidelinesNo	1.1117446	0.9200355	1.343400	0.2726715	1. (0 1.

▼ Code

```
# GGT
GGT.exercise.OR = bind_rows(extract_exercise_OR(GGT.crude, "Cru
                            extract_exercise_OR(GGT.main,  "Adj
GGT.exercise.OR = GGT.exercise.OR %>%
  mutate(`OR (95% CI)` = sprintf("%.2f (%.2f, %.2f)", OR, conf.
         `p-value` = ifelse(p.value < 0.0001, "<0.0001", sprint
kable(GGT.exercise.OR,
      caption = "Exercise Guidelines and GGT Category (Odds Rat
```

Exercise Guidelines and GGT Category (Odds Ratios)

Model	Outcome	term	OR	conf.low	conf.high	p.value	OI (95 CI
Crude	Borderline	ExerciseGuidelinesNo	1.274826	0.9949428	1.633442	0.0548736	1.2 (0.9 1.6
Crude	High	ExerciseGuidelinesNo	1.709266	1.4393842	2.029751	0.0000000	1.7 (1.4 2.0
Adjusted	Borderline	ExerciseGuidelinesNo	1.134011	0.8754303	1.468971	0.3408833	1.1 (0.8

							1.4
Adjusted	High	ExerciseGuidelinesNo	1.349786	1.1229928	1.622380	0.0013939	1.3 (1.1 1.6

The interaction table is drafted below, if needed.

▼ Code

```
# Interactions
extract_interactions = function(model) {

  tidy(model) %>% filter(grepl("^ExerciseGuidelines", term)) %>%
    select(Outcome = y.level, term, estimate, std.error, p.value

}

ALT.interactions = extract_interactions(ALT.int)
GGT.interactions = extract_interactions(GGT.int)

kable(ALT.interactions, digits = 3,
      caption = "ALT Interaction Model: Exercise-Related Terms")
```

ALT Interaction Model: Exercise-Related Terms

Outcome	term	estimate	std.error	p.value
Borderline	ExerciseGuidelinesNo	0.015	0.139	0.914
Borderline	ExerciseGuidelinesNo:AlcoholRiskMedium/High risk	-0.027	0.289	0.925
Borderline	ExerciseGuidelinesNo:SmokingDurationLess than 10 years	-0.423	0.307	0.168
Borderline	ExerciseGuidelinesNo:SmokingDuration10 to 19 years	0.053	0.304	0.861
Borderline	ExerciseGuidelinesNo:SmokingDuration20 to 29 years	-0.408	0.334	0.221
Borderline	ExerciseGuidelinesNo:SmokingDuration30 to 39 years	-0.934	0.406	0.021
Borderline	ExerciseGuidelinesNo:SmokingDuration40 to 49 years	0.427	0.500	0.393
Borderline	ExerciseGuidelinesNo:SmokingDuration50 years and over	0.753	1.150	0.512

High	ExerciseGuidelinesNo	0.235	0.140	0.093
High	ExerciseGuidelinesNo:AlcoholRiskMedium/High risk	-0.229	0.263	0.383
High	ExerciseGuidelinesNo:SmokingDurationLess than 10 years	0.387	0.326	0.234
High	ExerciseGuidelinesNo:SmokingDuration10 to 19 years	-0.298	0.288	0.302
High	ExerciseGuidelinesNo:SmokingDuration20 to 29 years	-0.377	0.308	0.221
High	ExerciseGuidelinesNo:SmokingDuration30 to 39 years	-0.037	0.353	0.918
High	ExerciseGuidelinesNo:SmokingDuration40 to 49 years	-0.703	0.406	0.083
High	ExerciseGuidelinesNo:SmokingDuration50 years and over	-0.378	0.936	0.687

▼ Code

```
kable(GGT.interactions, digits = 3,
      caption = "GGT Interaction Model: Exercise-Related Terms")
```

GGT Interaction Model: Exercise-Related Terms

Outcome	term	estimate	std.error	p.value
Borderline	ExerciseGuidelinesNo	0.179	0.202	0.376
Borderline	ExerciseGuidelinesNo:AlcoholRiskMedium/High risk	-0.611	0.341	0.073
Borderline	ExerciseGuidelinesNo:SmokingDurationLess than 10 years	0.093	0.404	0.819
Borderline	ExerciseGuidelinesNo:SmokingDuration10 to 19 years	0.049	0.378	0.896
Borderline	ExerciseGuidelinesNo:SmokingDuration20 to 29 years	0.453	0.432	0.294
Borderline	ExerciseGuidelinesNo:SmokingDuration30 to 39 years	-0.273	0.523	0.601
Borderline	ExerciseGuidelinesNo:SmokingDuration40 to 49 years	0.326	0.608	0.593
Borderline	ExerciseGuidelinesNo:SmokingDuration50 years and over	-0.277	0.731	0.705
High	ExerciseGuidelinesNo	0.314	0.141	0.026

High	ExerciseGuidelinesNo:AlcoholRiskMedium/High risk	-0.231	0.233	0.323
High	ExerciseGuidelinesNo:SmokingDurationLess than 10 years	-0.149	0.320	0.642
High	ExerciseGuidelinesNo:SmokingDuration10 to 19 years	0.226	0.306	0.460
High	ExerciseGuidelinesNo:SmokingDuration20 to 29 years	0.019	0.286	0.948
High	ExerciseGuidelinesNo:SmokingDuration30 to 39 years	-0.117	0.319	0.713
High	ExerciseGuidelinesNo:SmokingDuration40 to 49 years	0.290	0.379	0.444
High	ExerciseGuidelinesNo:SmokingDuration50 years and over	0.271	0.670	0.686

Plots

Plots 1: `ExerciseGuidelines` with `ALT3/GGT3`

To use the categorical variables as in the main model, we will set the y-axis as the predicted probabilities instead of the numerical values. Also the error bars are using 95% confidence intervals.

▼ Code

```
# ALT predicted probabilities
ALT.prob = emmeans(ALT.main, ~ ALT3 | ExerciseGuidelines, mode =
  as.data.frame() %>% rename(Outcome = ALT3) %>% mutate(Biomarke

# GGT predicted probabilities
GGT.prob = emmeans(GGT.main, ~ GGT3 | ExerciseGuidelines, mode =
  as.data.frame() %>% rename(Outcome = GGT3) %>% mutate(Biomarke

# Combine
biomarker.prob = bind_rows(ALT.prob, GGT.prob)
```

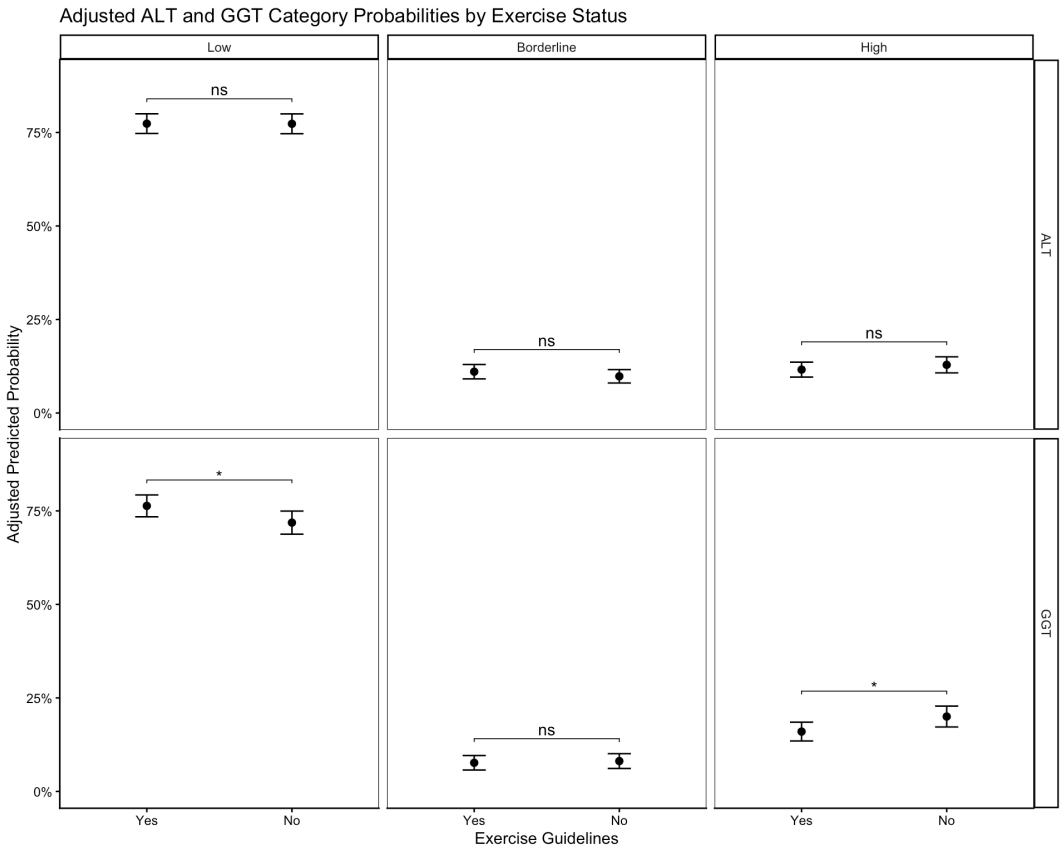
▼ Code

```
# Significance tests
ALT.exercise.test = emmeans(ALT.main, ~ ExerciseGuidelines | AL
                            mode = "prob") %>%
  pairs(adjust = "none") %>% test(by = NULL, adjust = "holm") %
  as.data.frame() %>% rename(Outcome = ALT3) %>% mutate(Biomark
GGT.exercise.test = emmeans(GGT.main, ~ ExerciseGuidelines | GG
                            mode = "prob") %>%
  pairs(adjust = "none") %>% test(by = NULL, adjust = "holm") %
  as.data.frame() %>% rename(Outcome = GGT3) %>% mutate(Biomark

exercise.tests = bind_rows(ALT.exercise.test, GGT.exercise.test
  mutate(signif = case_when(p.value < 0.0001 ~ "****",
                            p.value < 0.001 ~ "***",
                            p.value < 0.01 ~ "**",
                            p.value < 0.05 ~ "*",
                            TRUE ~ "ns"))

annotation.dat = biomarker.prob %>% group_by(Biomarker, Outcome
  summarise(y.position = max(upper.CL) + 0.04, .groups = "drop"
  left_join(exercise.tests %>% select(Biomarker, Outcome, signi
            by = c("Biomarker", "Outcome")) %>%
  mutate(group1 = "Yes", group2 = "No")

prob.plot = ggplot(biomarker.prob, aes(x = ExerciseGuidelines, y
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0
  stat_pvalue_manual(annotation.dat, xmin = "group1", xmax = "gr
                     y.position = "y.position", label = "signif"
                     tip.length = 0.01) +
  facet_grid(Biomarker ~ Outcome) +
  scale_y_continuous(labels = scales::percent_format(accuracy =
                     limits = c(0, 0.9)) +
  labs(x = "Exercise Guidelines", y = "Adjusted Predicted Probab
       title = "Adjusted ALT and GGT Category Probabilities by E
  theme_classic() + theme(panel.border = element_rect(colour = "
                                                      fill = NA,
                                                      linewidth
prob.plot
```



Contrasts and Difference Plots

Some numerical results and tests as well for the contrasts; these test within each ALT/GGT category, how the model-adjusted predicted probability differ between `ExerciseGuidelines` is Yes and No. These are not tests of the multinomial coefficients themselves, but are a better representation of the significance stars in the plot.

▼ Code

```
# ALT
ALT.exercise.test = emmeans(ALT.main, ~ ExerciseGuidelines | AL
                            mode = "prob") %>% pairs(adjust = "
ALT.exercise.test
```

```
ALT3 = Low:
 contrast  estimate     SE df t.ratio p.value
 Yes - No  0.000405 0.0129 44   0.032  0.9750

ALT3 = Borderline:
 contrast  estimate     SE df t.ratio p.value
 Yes - No  0.012353 0.0090 44   1.373  0.1766
```

```
ALT3 = High:
 contrast  estimate     SE df t.ratio p.value
 Yes - No -0.012758 0.0101 44  -1.266  0.2122

Results are averaged over the levels of: Sex, AlcoholRisk,
SmokingDuration, NonSchoolQuals, IncomeDecile
```

▼ Code

```
# GGT
GGT.exercise.test = emmeans(GGT.main, ~ ExerciseGuidelines | GG
                            mode = "prob") %>% pairs(adjust = "
GGT.exercise.test
```

```
GGT3 = Low:
 contrast estimate      SE df t.ratio p.value
 Yes - No  0.04478 0.01470 44   3.055  0.0038

GGT3 = Borderline:
 contrast estimate      SE df t.ratio p.value
 Yes - No -0.00455 0.00939 44  -0.485  0.6303

GGT3 = High:
 contrast estimate      SE df t.ratio p.value
 Yes - No -0.04023 0.01310 44  -3.069  0.0037

Results are averaged over the levels of: Sex, AlcoholRisk,
SmokingDuration, NonSchoolQuals, IncomeDecile
```

Interpretation.

- Low ALT: Yes - No = 0.000405, or about +0.04 percentage points. Essentially no difference; $p = 0.9750$.
- Borderline ALT: Yes - No = 0.012353, or about +1.24 percentage points for people meeting the exercise guideline. $p = 0.1766$, so no statistically significant evidence of a difference.
- High ALT: Yes - No = -0.012758, or about −1.28 percentage points. People meeting the guideline have a slightly lower adjusted predicted probability of High ALT, but $p = 0.2122$, so again there is no statistically significant evidence of a difference.
- Overall: after adjustment for the covariates in ALT.main, there is no clear evidence that ALT-category probabilities differ by exercise-guideline status.

- Low GGT: Yes - No = 0.04478, so people meeting the exercise guideline have an adjusted predicted probability of Low GGT about 4.48 percentage points higher than those not meeting it. $p = 0.0038$.
- Borderline GGT: difference = -0.00455, or about −0.46 percentage points. This is tiny and not statistically significant, $p = 0.6303$.
- High GGT: difference = -0.04023, meaning people meeting the exercise guideline have an adjusted predicted probability of High GGT about 4.02 percentage points lower. $p = 0.0037$.
- Overall: the model suggests exercise-guideline status is associated mainly with a shift from the High GGT category toward the Low GGT category, rather than much change in the Borderline category.

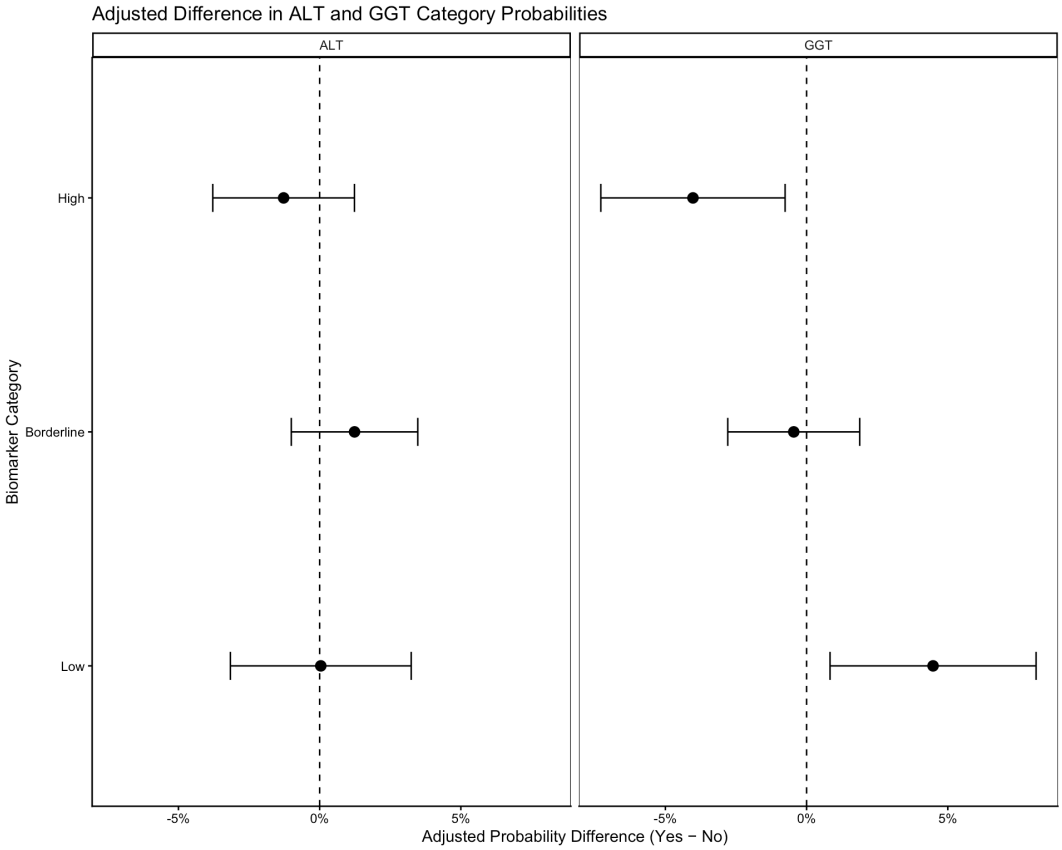
We can make a related difference plot for these as well, which may be better for us compared to the previous plot for interpretations.

▼ Code

```
# Differences
ALT.diff = emmeans(ALT.main, ~ ExerciseGuidelines | ALT3, mode =
  pairs(adjust = "holm") %>% confint(by = NULL, adjust = "holm")
  as.data.frame() %>% rename(Outcome = ALT3) %>% mutate(Biomarke

GGT.diff = emmeans(GGT.main, ~ ExerciseGuidelines | GGT3, mode =
  pairs(adjust = "holm") %>% confint(by = NULL, adjust = "holm")
  as.data.frame() %>% rename(Outcome = GGT3) %>% mutate(Biomarke
prob.diff = bind_rows(ALT.diff, GGT.diff)

# Plot
diffs = ggplot(prob.diff, aes(x = estimate, y = Outcome)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = lower.CL, xmax = upper.CL), height =
  facet_wrap(~ Biomarker) +
  scale_x_continuous(labels = scales::percent_format(accuracy =
  labs(x = "Adjusted Probability Difference (Yes - No)",
       y = "Biomarker Category",
       title = "Adjusted Difference in ALT and GGT Category Prob
  theme_classic() +
  theme(panel.border = element_rect(colour = "black", fill = NA,
diffs
```



Interaction Plots

We plot the interaction difference plots e.g., plotting the value of

$$P(\text{High} \mid \text{Exercise}{=}\text{Yes}) - P(\text{High} \mid \text{Exercise}{=}\text{No})$$

for each set of relevant variables.

▼ Code

```
# Helper function
get_interaction_diff = function(model, outcome, moderator, bioma

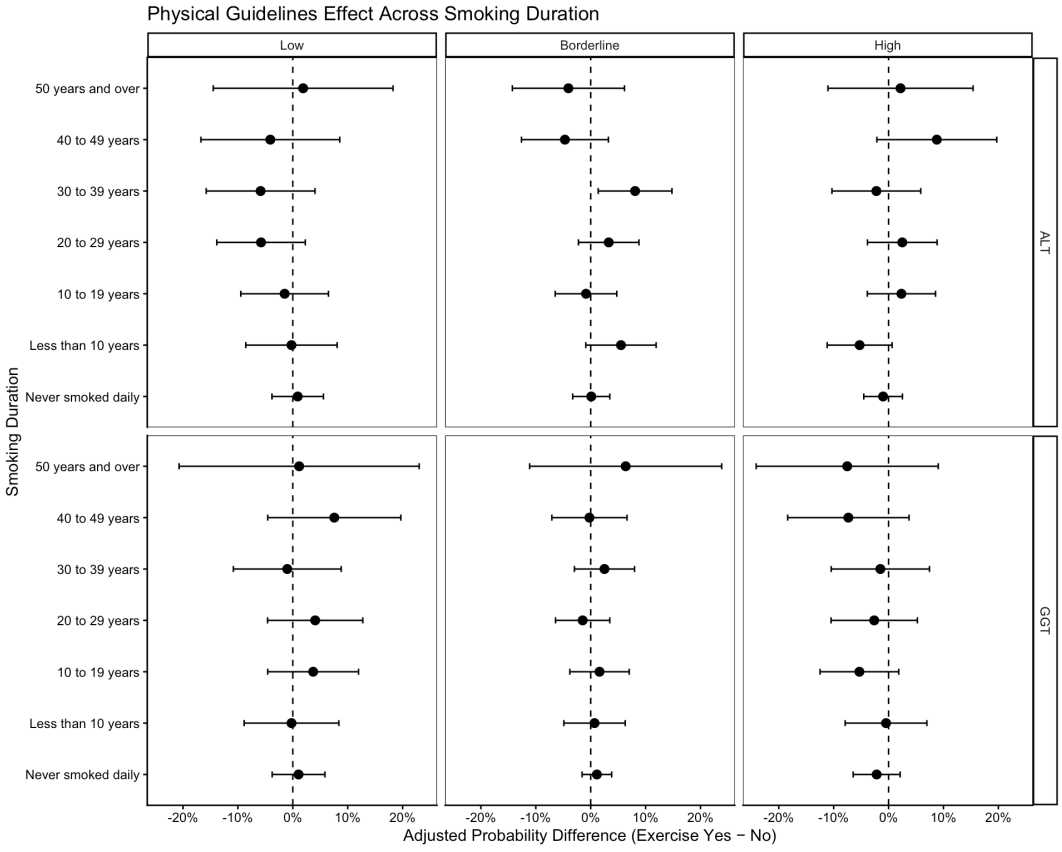
  specs = as.formula(paste("~ ExerciseGuidelines |", moderator,
  emm = emmeans(model, specs, mode = "prob")

  pairs(emm, adjust = "none") %>%
    summary(infer = c(TRUE, TRUE), level = 0.95, adjust = "none'
    as.data.frame() %>%
    rename(Level = all_of(moderator), Outcome = all_of(outcome)
    mutate(Biomarker = biomarker)

}
```

```
# Smoking
ALT.smoking.diff = get_interaction_diff(ALT.int, outcome = "ALT
                                        moderator = "SmokingDura
                                        biomarker = "ALT")
GGT.smoking.diff = get_interaction_diff(GGT.int, outcome = "GGT
                                        moderator = "SmokingDura
                                        biomarker = "GGT")
smoking.diff = bind_rows(ALT.smoking.diff, GGT.smoking.diff)

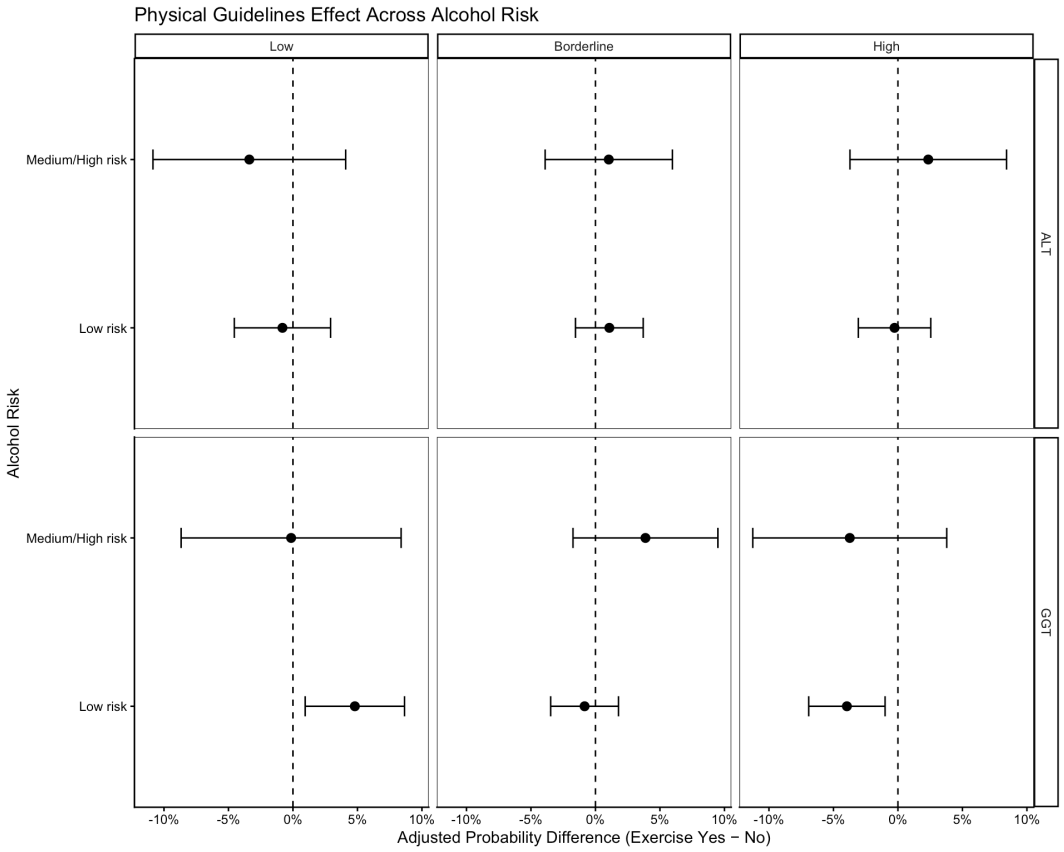
smoking = ggplot(smoking.diff, aes(x = estimate, y = Level)) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.
  geom_errorbarh(aes(xmin = lower.CL, xmax = upper.CL), height =
  geom_point(size = 2.5) + facet_grid(Biomarker ~ Outcome) +
  scale_x_continuous(labels = percent_format(accuracy = 1)) +
  labs(x = "Adjusted Probability Difference (Exercise Yes - No)"
       y = "Smoking Duration",
       title = "Physical Guidelines Effect Across Smoking Durati
  theme_classic() +
  theme(panel.border = element_rect(colour = "black", fill = NA,
smoking
```



▼ Code

```
# Alcohol
ALT.alcohol.diff = get_interaction_diff(ALT.int,outcome = "ALT3'
                                        moderator = "AlcoholRisl
                                        biomarker = "ALT")
GGT.alcohol.diff = get_interaction_diff(GGT.int, outcome = "GGT3
                                        moderator = "AlcoholRisl
                                        biomarker = "GGT")
alcohol.diff = bind_rows(ALT.alcohol.diff, GGT.alcohol.diff)

# Plot
alc = ggplot(alcohol.diff, aes(x = estimate, y = Level)) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.
  geom_errorbarh(aes(xmin = lower.CL, xmax = upper.CL), height =
  geom_point(size = 2.5) + facet_grid(Biomarker ~ Outcome) +
  scale_x_continuous(labels = percent_format(accuracy = 1)) +
  labs(x = "Adjusted Probability Difference (Exercise Yes - No)"
       y = "Alcohol Risk",
       title = "Physical Guidelines Effect Across Alcohol Risk")
  theme_classic() +
  theme(panel.border = element_rect(colour = "black", fill = NA,
alc
```



Note that the nteraction model/test provides evidence about whether effect modification existsl the interaction-difference plots above tells us where that

modification appears to occur and how large the subgroup-specific differences are.